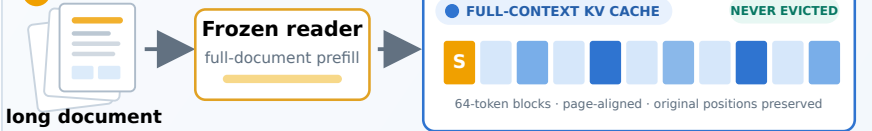


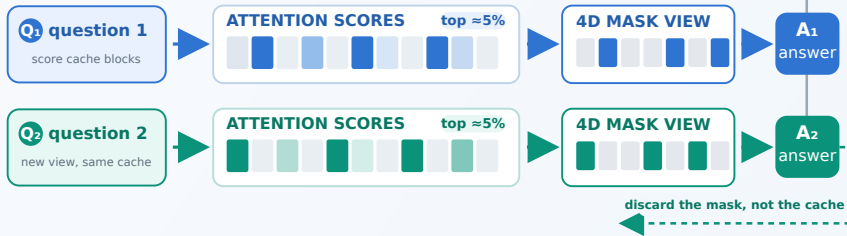
a PAQ-KV: prefill once, ask many

One intact cache · a new evidence view for every question

1 ONE-TIME PREFILL



2 RE-SELECT PER QUERY 3 MASKED DECODE IN PLACE



NO GATHER · NO EVICTION · NO RE-READ · 0 EXTRA PARAMETERS

b Sealed evaluation

MMLongBench-Doc

fresh, disjoint test set

matched nominal 5% evidence budget

BENCHMARK ACCURACY **+0.076**

PAQ-KV

0.396

ColQwen2

0.320

✓ 95% CI [+0.029, +0.122]

document-cluster paired bootstrap

45 documents · 244 / 244 queries